Week 9

Introduction

In the dynamic landscape of power electronics, where efficiency, reliability, and precise control are paramount, understanding the intricate behaviors of DC-DC converters is essential. This week, we delve into the sophisticated world of power conversion, focusing on the critical techniques that enable the design and optimization of high-performance systems, particularly relevant for applications like electric vehicle charging infrastructure. We begin by exploring small signal modeling, a fundamental analytical tool that linearizes complex converter dynamics, allowing engineers to predict behavior and design robust control systems. This foundation then leads us to the **Dual Active Bridge Converter**, a cutting-edge topology renowned for its bidirectional power flow capabilities and high power density, addressing many limitations of its predecessors. Finally, we tackle the crucial aspect of high-frequency transformer design, unraveling the complexities of magnetic materials, winding configurations, and loss mechanisms that dictate the overall efficiency and compactness of isolated power converters. Together, these topics provide a comprehensive understanding of how to engineer power solutions that are not only efficient but also stable, controllable, and tailored for the demanding requirements of modern power systems.

Core Concepts

Term	Definition	Significance
`GVD(S)`	The transfer function relating small signal output voltage variation to small signal duty ratio variation, with constant input voltage.	Crucial for designing voltage controllers, quantifying how output voltage responds to control input.
`Dual Active Bridge (DAB)`	An isolated DC-DC converter with two active full-bridge circuits, one on each side of a high-frequency transformer.	Enables bidirectional power flow and offers superior control flexibility for high-power applications.
`Single Phase Shift (SPS) Modulation`	A DAB control strategy where power flow is regulated by varying the phase shift between primary and secondary bridge voltages.	The simplest modulation scheme, offering fundamental control over power direction and magnitude.
`Flux Walking`	An undesirable phenomenon where the magnetic flux in a transformer	Can cause transformer saturation and failure; requires

Term	Definition	Significance
	core accumulates a DC bias, potentially leading to saturation.	control strategies like peak current mode control to mitigate.
`Core Loss`	Energy dissipated as heat within the transformer's magnetic core due to alternating magnetic fields (hysteresis and eddy currents).	A major contributor to overall transformer inefficiency and heating, requiring careful design to minimize.
`Peak Flux Density (ΔB)`	The peak-to-peak change in AC magnetic flux density within the transformer core during operation.	A key design parameter that directly influences core losses and must be kept below the saturation flux density.
`Window Area (WA)`	The physical space within the transformer core structure available for accommodating the windings.	Must be sufficiently large to house the required number of turns and wire size, impacting transformer dimensions.
`Winding Fill Factor (Ku)`	A dimensionless factor representing the ratio of the actual copper area to the total available window area.	Accounts for non-ideal packing of wires and insulation, indicating how effectively the window area is utilized.

Detailed Analysis

1. Small Signal Modeling of PSFB Converters

The week began with an in-depth look at **Small Signal Modelling of PSFB-III**, a critical technique for analyzing the dynamic behavior of power converters. Small signal modeling linearizes the converter's complex switching behavior around a specific operating point, allowing for the derivation of transfer functions essential for controller design. The primary focus was on deriving `GVD(S)`, the transfer function relating the small signal output voltage variation to the small signal duty ratio variation, assuming a constant input voltage.

The derivation of `GVD(S)` involves several intermediate steps:
- **V_AB and H0(S) Derivation**: The small signal voltage across the transformer's secondary, `V_AB_tilde`, is expressed as `n * V_in * d_tilde`. The transfer function `H0(S)` then describes the dynamics of the output LC filter and load, relating `V_out(S)` to `V_AB(S)`.

- **Input Impedance (Z_in_AB)**: This represents the impedance seen looking into the output filter from the transformer secondary, encompassing the inductor `L`, capacitor `C`, and load `R_L`.
- **DI and GVD(S) Derivation**: The small signal current variation `DI` is related to `d_tilde` and `V_in` through an equivalent resistance `R_equivalent`, which accounts for leakage inductance and switching frequency. By combining `H0(S)`, `V_AB(S)`, and the relationship involving `DI(S)`, the comprehensive `GVD(S)` transfer function is obtained.

With `GVD(S)` in hand, the design of a closed-loop control system becomes feasible. A typical system includes a reference voltage, feedback voltage, error signal, a controller `C(S)`, a gain block `K`, a modulator block, the PSFB converter (`GVD(S)`), and a sensor `H(S)`. Controller design guidelines emphasize high gain at DC for zero steady-state error, a sufficient phase margin (typically >35 degrees) for stability, and an appropriate bandwidth for effective disturbance rejection. Practical implementations often involve microcontrollers for `C(S)` and `K`, and advanced considerations like peak current mode control are used to prevent transformer "flux walking" and saturation.

Introduction

Building on the understanding of PSFB converters, the lecture on **Dual Active Bridge Converter-I** introduced a more advanced topology designed to overcome the PSFB's limitations. The PSFB suffered from unidirectional power flow, limited soft-switching range, large secondary inductive filters, and susceptibility to flux walking.

The DAB converter addresses these issues with its unique structure:
- **Two Active Bridges**: It features a full-bridge inverter on the primary side and another full-bridge inverter/rectifier on the secondary side, connected by a high-frequency transformer and a series inductor `L`.
- **Generic Switches**: The use of generic switches (e.g., MOSFETs with body diodes) allows for bidirectional current flow.
- **Inductor Role**: The series inductor `L` is crucial for energy transfer, enabling power flow control by applying different voltages across it from the primary and secondary bridges.

The DAB offers significant control flexibility through multiple variables:
- `Delta (δ)`: Phase shift between the primary and secondary full-bridge voltages.
- `Theta_1 (θ1)`: Phase shift within the primary full-bridge.
- `Theta_2 (θ2)`: Phase shift within the secondary full-bridge.

Various modulation schemes arise from these variables:
- **Single Phase Shift (SPS) Modulation**: Only `delta` is varied, offering the simplest control.
- **Dual Phase Shift (DPS) Modulation**: Varies `delta` and either `theta_1` or `theta_2`.
- **Triple Phase Shift (TPS) Modulation**: Controls `delta`, `theta_1`, and `theta_2` for maximum flexibility.

- **Variable Frequency (VF) Modulation**: Can be combined with any phase shift scheme to alter inductor impedance.

Waveform analysis for SPS modulation (where `theta_1` and `theta_2` are zero) shows square wave voltages `VA1B1` and `VA2B2` (referred to primary), with `VA2B2` phase-shifted by `delta`. The inductor current `IL` waveform is piecewise linear, with slopes determined by the voltage difference across `L`. Symmetrical current waveforms are essential to prevent flux walking.

3. Advanced DAB Operation and Power Transfer

The discussion continued with **Dual Active Bridge Converter-II**, delving deeper into DAB modulation schemes and power transfer analysis. The various modulation techniques (SPS, DPS, TPS, and VF) were further elaborated, highlighting their impact on current waveforms and efficiency.

A detailed analysis of the inductor current (`IL`) waveform for SPS modulation was presented, particularly for the case where `V_in > V_out/N`. The current waveform is piecewise linear, with slopes determined by the voltage across the inductor. The average input current and power transfer (`P_in_average`) were derived:
`P_in_average = V_in^2 * δ* * (1 - δ*) / (2 * N * L * FSW)`
where `δ*` is the normalized phase shift (`delta / 180°`). This equation reveals that maximum power transfer occurs when `δ* = 0.5` (i.e., `delta = 90°`), resulting in `P_average_maximum = V_in^2 / (8 * N * L * FSW)`. This parabolic power characteristic is crucial for control.

The selection of the series inductance `L` is critical, constrained by the maximum desired power transfer at minimum voltage conditions. The lecture also covered:
- **Backward Power Flow**: Achieved by applying a leading phase shift to the secondary bridge voltage, reversing the power direction.
- **Impact of Modulation on RMS Current**: Different modulation schemes lead to varying RMS currents, directly affecting conduction losses and thermal management. DPS and TPS can often reduce RMS currents compared to SPS.
- **Advanced DAB Topologies**: Resonant DABs use LC networks to achieve soft switching and filter harmonics, while Multi-level DABs use multi-level half-bridges to reduce voltage stress and RMS currents, improving efficiency and power quality.

4. Fundamentals of High-Frequency Transformer Design

The focus then shifted to **Transformer Design for Isolated DC-DC Converter-I**, emphasizing the unique challenges and considerations for high-frequency transformers in isolated DC-DC converters. Unlike traditional 50/60 Hz transformers, these operate at much higher frequencies (tens of kHz to MHz), impacting their size, efficiency, and material selection.

Key concepts covered included:
- **Magnetic Circuit Model**: Explaining how magnetizing current generates flux in the core, relating magnetic field strength `H` to turns `N1` and magnetic path length `LC`.
- **BH Curve and Saturation**: The BH curve illustrates the relationship between magnetic flux density `B` and magnetic field strength `H`. Understanding magnetic domains and the critical point of magnetic saturation is vital, as operating beyond saturation leads to significant losses and performance degradation.
- **Faraday's Law and Flux Linkage**: The induced voltage is proportional to the rate of change of flux linkage. The peak AC flux density (`ΔB`) is derived from the applied voltage and must be kept below the core's saturation flux density (`B_sat`).
- **Transformer Design Outcomes**: The goal is to determine core size (cross-sectional area `AC`, window area `WA`, magnetic path length `LC`), number of turns, wire size, and the optimal `ΔB`.
- **Core Losses**: Comprising hysteresis loss (energy to reorient magnetic domains) and eddy current loss (circulating currents induced by changing flux). Hysteresis loss is proportional to frequency and `ΔB^α`, while eddy current loss is proportional to `f^2` and `ΔB^2`. The Empirical Steinmetz Formula (`P_core = K_core * ΔB^β * Volume`) is used to estimate total core losses from manufacturer data.

5. Optimizing Transformer Performance

The final lesson, **Transformer Design for Isolated DC-DC Converter-II**, delved into the intricate details of optimizing transformer performance by minimizing total losses. The design process involves balancing core losses and winding (copper) losses.

Key aspects of optimization include:
- **Winding Losses (Copper Losses)**: Calculated as `I^2 * R`, where `R` is the winding resistance. The resistance `R_j` for winding `j` depends on wire resistivity `ρ`, total wire length `L_j`, and wire cross-sectional area `A_w,j`.
- **Window Area (WA) and Winding Allocation (`γ_j`)**: The `WA` is the total space for windings. Each winding `j` is allocated a fraction `γ_j` of this area, where the sum of all `γ_j` must be 1.
- **Mean Length Per Turn (MLT) and Winding Fill Factor (Ku)**: `MLT` is the average length of a single turn, while `K_u` accounts for non-ideal wire packing and insulation (typically < 1). These factors are crucial for accurately calculating `L_j` and `A_w,j`.
- **Total Copper Loss Calculation**: The total copper loss is derived as `P_cu_total = (ρ * MLT * N_1^2 / (WA * K_u)) * Σ ((N_j/N_1)^2 * I_j^2 / γ_j)`.
- **Two-Level Optimization Strategy**:
1. **Gamma Optimization**: Minimizing total copper loss by finding the optimum `γ_j` for each winding. This ensures efficient utilization of the window area.
2. **Delta B Optimization**: Minimizing the overall total loss (`P_total = P_core + P_cu_total`) by finding the optimum `ΔB`. `P_core` increases with `ΔB`, while `P_cu_total`

decreases with `ΔB` (due to fewer turns). This results in a U-shaped total loss curve, indicating an optimal `ΔB` where losses are minimal.

This systematic approach allows engineers to determine the optimal core size, number of turns, and wire size for a high-frequency transformer, ensuring efficient and reliable operation.

Key Takeaways

- [ ] Small signal modeling linearizes converter behavior to derive transfer functions like `GVD(S)`, crucial for control system design.
- [ ] The **Dual Active Bridge (DAB) Converter** enables bidirectional power flow and offers superior control flexibility compared to PSFB converters.
- [ ] **Single Phase Shift (SPS) Modulation** is the simplest DAB control, regulating power by varying the phase shift (`delta`) between primary and secondary bridges.
- [ ] Maximum power transfer in SPS modulation occurs when the normalized phase shift `delta_star` is 0.5 (i.e., `delta = 90°`).
- [ ] Transformer core losses (hysteresis and eddy currents) and winding losses (copper losses) are the primary contributors to inefficiency and heat generation.
- [ ] The **Empirical Steinmetz Formula** is used to estimate total core losses, which are highly dependent on **`Peak Flux Density (ΔB)`** and frequency.
- [ ] Magnetic saturation must be avoided in transformers, and `ΔB` must be carefully chosen to minimize losses while staying below `B_sat`.
- [ ] Transformer design involves a two-level optimization: first, allocating **`Window Area (WA)`** fractions (`Y_j`) to minimize copper losses, then optimizing `ΔB` to minimize total losses.
- [ ] **`Flux Walking`** in transformers can lead to saturation and must be prevented through careful control strategies, such as peak current mode control.

Conclusion

This week's exploration has equipped us with a powerful toolkit for designing and optimizing advanced power electronic converters. From the analytical rigor of small signal modeling, which provides the blueprint for stable control systems, to the innovative architecture of the **Dual Active Bridge Converter** enabling bidirectional power flow, we've seen how theoretical understanding translates into practical capabilities. The detailed dive into high-frequency transformer design, encompassing core material characteristics, loss mechanisms, and sophisticated optimization strategies, underscores the critical role of every component in achieving overall system efficiency and reliability. These concepts are not merely academic exercises; they are the foundational principles driving the development of next-generation power solutions for electric vehicle charging, renewable energy integration, and robust industrial power supplies. As we move forward, the ability to precisely model, control, and

optimize these complex systems will be indispensable for engineers shaping the future of power electronics.